

NANDHAKUMAR G

MECHANICAL ENGINEERING UNDERGRADUATE (2027 BATCH)

Analytical and detail-oriented Mechanical Engineering student with hands-on industrial traineeship experience in heavy manufacturing machinery, raw material handling, and steel processing. Proficient in engineering drawing interpretation and CAD assemblies design verification, with dynamic experience in electromechanical integrations (ROS, sensor nodes, and microcontrollers). Skilled in compiling comprehensive technical reports and structured layout documentations.

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SCAN TO VIEW PORTFOLIO

TECHNICAL FOCUS

- ✓ **Drawing & Schemes:** Strong capabilities in reading, analyzing, and verifying engineering blueprints and electrical schematics layouts.
- ✓ **System Documentation:** Experienced in drafting structured mechanical specifications, parts manuals, assembly processes, and validation checklists.
- ✓ **Diagnostics Support:** Quick to troubleshoot electromechanical systems, control loops, and component assemblies clearances.

CORE COMPETENCIES

TECHNICAL SUPPORT & DOCS

- Drawing Reading
- Technical Writing
- Parts Manuals
- Component Diagrams

CAD PROTOTYPING & DFM

- 3D Gearing Assemblies
- Tolerance Stacks
- DFM Principles
- Stress Analysis

KEY ATTRIBUTES

Analytical Mindset:

Methodical approach to tracing machine errors and electrical wiring malfunctions.

Clear Communication:

Adept at explaining complex physical parts layouts and workflows in writing.

Continuous Learner:

Passion for discovering mechanical operations and adapting to technical diagnostics tools.

TECHNICAL TOOLS & SOFTWARE

Fusion 360	SolidWorks	Siemens NX	AutoCAD
Python	C++	MATLAB	COMSOL

PROFESSIONAL EXPERIENCE

Industrial Engineering Trainee Steel Authority of India Ltd. (SAIL), Salem Steel Plant

Dec 2025

- Inspected operations of integrated hot rolling steel mill structures, hydraulics control systems, and heavy casting assemblies.
- Read and interpreted complex mill blueprints, piping layout schematics, and structural drawings.
- Documented maintenance protocols, equipment operational schedules, and machinery safety checklists.
- Collaborated with senior plant technicians to evaluate machinery performance parameters and wear indicators.

Reverse Engineering Intern Femlogic Industries, Chennai

Jun 2025 - Jul 2025

- Performed reverse engineering analysis and parametric 3D CAD modeling for Vande Bharat passenger seating structural interfaces.
- Analyzed physical models and clearance tolerances, converting components geometry into standard manufacturing layouts.
- Drafted structural tolerance stacks and mechanical interface reports for parts verification engineering teams.
- Applied Design for Manufacturing (DFM) guidelines to optimize fasteners selections and thin sheet metal stampings.

Production Engineering Intern K.M. Knitwear Pvt. Ltd., Tirupur

Jul 2024 - Aug 2024

- Analyzed flow diagrams of floor packaging machinery, creating recommendations that boosted rates output by 10%.
- Drafted structural design layouts for materials storage components to secure items during shipping cycles.
- Participated in mechanical troubleshooting and diagnostics checks on machinery drive linkages.

SELECTIONS

SELECTED ENGINEERING PROJECTS

Project EcoFloat | Niral Hackathon Award Winner

SYSTEMS LEAD & DESIGNER

ROS · IoT · Fusion 360 · Arduino

> Designed and co-developed an autonomous twin-hull catamaran robot for floating plastic waste collection and real-time chemical water diagnostics.

> Configured sensor node communications (serial ports interface) and path tracking scripts integrated with ROS edge nodes.

> Designed the full physical parametric assembly in Fusion 360, selecting materials for structural load, center of mass, and buoyancy calculations.

> Compiled all system technical reports, physical assembly diagrams, parts manuals, and standard operational documentation.

Hemispherical CPV Solar Tracking System

MECHANICAL DESIGNER

COMSOL · Solar Math · Fusion 360

> Modeled a compound dual-axis Concentrator Photovoltaic solar tracking dome structure designed to maximize solar vector energy absorption.

> Conducted finite element (FEA) thermal and structural safety factor simulations in COMSOL to validate high wind-load tolerances.

> Drafted mechanical blueprints, assembly calculations, and mathematical solar tracking formulas.

EV Regenerative Braking Control Loop

DEVELOPER & RESEARCHER

MATLAB Simulink · Powertrain Modeling

> Designed an active kinetic-to-electrical energy recovery control loop for electric vehicle prototypes using MATLAB Simulink.

> Integrated battery management system (BMS) controls to monitor thermal state-of-charge curves, extending battery lifecycle metrics by 20%.

> Compiled comprehensive simulation test reports, transient response graphs, and component specifications registers.

EDUCATION QUALIFICATIONS

Bachelor of Mechanical Engineering
(2027 Batch)

2023 - Present

St. Joseph's College of Engineering, Chennai,
Tamil Nadu

Core Coursework: Thermodynamics, Fluid
Mechanics, Machine Design, Control Systems

Higher Secondary Education (Class XII)

2022 - 2023

Shri Swamy Matriculation Higher Secondary
School, Salem, Tamil Nadu

Computer Science & Mathematics Stream

Secondary School Education (Class X)

2020 - 2021

Shri Swamy Matriculation Higher Secondary
School, Salem, Tamil Nadu

Matriculation State Board Curriculum

PROFESSIONAL CREDENTIALS

Autodesk Certified Fusion Designer

Professional CAD Design Credential

Certified Solidworks Designer

Mechanical Assemblies & DFM Spec

Siemens NX CAD Designer

Advanced Parametric Surfaces Spec

AutoCAD for 2D Modelling

Technical Drafting & Detailing Standard

Certified in Python Advanced

Computational Data Structures

C++ Programming

OOP & Embedded Software Logic

LANGUAGES

English

Fluent (Written & Verbal)

Tamil

Native Speaker